

LOUCA MALERBA

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EXPERIENCE

GE HealthCare, Buc, France

June 2025 – Present

Apprentice Research Engineer

- Developing deep learning models for automated detection and segmentation of chronic pulmonary embolisms on 3D CT scans (nnU-Net, nnDetection, PyTorch).
- Embedded in a larger research program conducted with Kremlin-Bicêtre University Hospital (AP-HP); extending a codebase initiated by a former PhD candidate, with a strong focus on robustness, reproducibility, and performance.
- Supervision and governance: autonomous work with weekly guidance from senior research scientists/engineers on methodology, architecture choices, and quality assurance.

GE HealthCare, Buc, France

Oct. 2023 – June. 2025

Machine Learning & Data Science Apprentice

- Built predictive models to optimize equipment durability; weekly mentoring sessions with a PhD supervisor to refine implementation and optimization; designed and automated a network of test benches and data processing tools, producing datasets.
- Eight-week internship in Hungary: developed genetic algorithms to optimize task distribution among technicians, improving operational efficiency.

Freelancing

2023

Freelance Software Developer

- Web scraping for multiple applications; IT consulting for small clients.

EDUCATION

CentraleSupélec

2023 – 2026

General Engineering Master's, specialization in Data Science.

Lycée Jules Ferry, Versailles

2021 – 2023

Classe préparatoire aux grandes écoles, PTSI – PT* track.

Institution des Chartreux, Lyon

2018 – 2021

French Baccalauréat in Physics-Chemistry and Mathematics, High Honors.

Cistercian College, Roscrea (Ireland)

2018 – 2019

Full academic year abroad (grade 11).

English certification: Cambridge Advanced Certificate (C1).

SKILLS

Programming: Python, C++, PHP, JavaScript, HTML, CSS, SQL

Data/ML: PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

Tools/OS: Linux, MATLAB, Power BI, Git, DICOM handling, experiment tracking

Languages: English (C1), Italian (B1), Spanish (B1)

INTERESTS

Magnum Association (CentraleSupélec wine society): member; involved in event organization.

Competitive Sports: Cycling (7 years) and skiing (5 years) in competition; endurance and team spirit.